

Chapter 7D

Hematologic Malignancies, Plasma Cell Disorders, Lymphomas, Leukemias, and Oncology Foundations

Original Medvale study chapter for graph-RAG extraction. Source files are used as coverage signals only; all explanations, tables, and diagrams are newly written.

Why this matters clinically

Hematologic malignancies sit at the intersection of internal medicine, pathology, emergency care, and oncology. The same patient can present as a primary-care screening question, an inpatient with pancytopenia, an emergency with tumor lysis syndrome, a nephrology consult for light-chain cast nephropathy, or a neurology consult for hyperviscosity. The goal of this chapter is not to memorize every chemotherapy regimen; it is to build illness scripts that connect a few recurring patterns: abnormal blood counts, abnormal proteins, abnormal lymph nodes, marrow failure, cytokine symptoms, and end-organ injury.

A practical framework starts with the compartment involved. Plasma cell disorders produce monoclonal immunoglobulin and cause CRAB-type end-organ injury. Lymphomas usually begin as abnormal lymph nodes or extranodal lymphoid tissue. Leukemias are marrow and blood diseases that commonly present with cytopenias, leukocytosis, blasts, infection, bleeding, bone pain, or organ infiltration. Myeloproliferative neoplasms overproduce mature myeloid elements; myelodysplastic syndromes produce ineffective, dysplastic cells and progressive cytopenias.

Core pattern recognition

Clinical clue	High-yield interpretation	Do next
Painless lymphadenopathy plus fever, night sweats, weight loss	Lymphoma until proven otherwise; B symptoms influence staging/prognosis	Excisional lymph node biopsy when feasible; CT/PET staging after tissue diagnosis
Bone pain, anemia, renal failure, high calcium, recurrent infections	Plasma cell neoplasm with end-organ injury	SPEP/UPEP, serum free light chains, calcium/creatinine/CBC, marrow assessment, skeletal imaging
Very high WBC with blasts, anemia, thrombocytopenia	Acute leukemia; marrow packed with immature cells	Urgent hematology, smear/flow cytometry, marrow biopsy, DIC/TLS evaluation
Leukocytosis with basophilia and splenomegaly	Think CML, especially if granulocytes span maturation stages	BCR-ABL1 testing
Older adult with lymphocytosis and smudge cells	CLL/SLL spectrum	Flow cytometry; treat only if symptomatic/progressive

Oncology foundations: prevention, staging, treatment language

The uploaded oncology pages begin with epidemiology, screening, staging, and broad treatment modalities before moving into plasma cell and lymphoid malignancies. Those foundations are useful because hematology-oncology questions often ask what the clinician should do first: screen, diagnose, stage, stabilize, or treat. Smoking remains the most important avoidable cancer risk factor across many malignancies. Viral causes include HPV for cervical and several oropharyngeal cancers, HBV/HCV for hepatocellular carcinoma, EBV for several lymphoid malignancies, HTLV-1 for adult T-cell leukemia/lymphoma, and HIV-related immune dysfunction for aggressive lymphomas.

Cancer screening should be interpreted as population-level prevention, not as a diagnostic test for symptomatic disease. A patient with a breast mass, iron-deficiency anemia, rectal bleeding, hemoptysis, unexplained weight loss, or pathologic fracture needs diagnostic evaluation, not routine screening. When a malignancy is diagnosed, staging asks three questions: how large or locally invasive is the primary tumor, are regional lymph nodes involved, and is there distant metastasis. For solid tumors this is summarized by TNM; for lymphomas, Ann Arbor-style anatomic staging and PET-based assessment are more common.

Term	Meaning in practice	Clinical trap
Screening	Testing asymptomatic people in a risk-defined population	Do not use screening intervals to dismiss symptoms

Term	Meaning in practice	Clinical trap
Diagnosis	Tissue or laboratory confirmation of a suspected disease	Imaging can suggest cancer, but tissue usually defines cancer type
Staging	Measures extent of disease after diagnosis	Staging should not delay emergency stabilization
Neoadjuvant therapy	Systemic or radiation therapy before definitive local therapy	Used to shrink tumors or improve surgical/radiation outcomes
Adjuvant therapy	Treatment after definitive local therapy	Targets microscopic residual disease and recurrence risk
Targeted therapy	Therapy directed at a driver pathway or surface marker	Marker testing often determines eligibility
Immunotherapy	Treatment that modulates immune recognition or attack	Immune toxicities can mimic autoimmune disease

Clinical anchors from current oncology references

NCI PDQ summaries emphasize that disease-specific treatment summaries are peer-reviewed, evidence-based resources, but they are not formal decision-making guidelines. For a study resource, this means diagnostic logic and emergency stabilization are safe to learn at a broad level, while exact drug sequencing should be checked against current disease-specific guidelines and local oncology protocols.

Approach to abnormal counts: marrow, blood, and tissue compartments

Most hematologic malignancy evaluations start with a CBC with differential, smear, reticulocyte count if anemia is present, chemistry panel, uric acid, LDH, phosphorus, calcium, coagulation tests when bleeding or acute leukemia is suspected, and careful review of symptoms. The smear is often the bridge between nonspecific CBC abnormalities and a directed diagnosis: blasts suggest acute leukemia; smudge cells suggest CLL; rouleaux suggests excess plasma protein; teardrop cells suggest marrow fibrosis or infiltration; schistocytes suggest microangiopathy; basophilia with a left-shifted granulocytic spectrum suggests CML.

Original diagnostic map: CBC abnormality to malignant pattern
Step 1: Determine the dominant abnormality - cytopenia, leukocytosis, lymphocytosis, thrombocytosis, erythrocytosis, or paraprotein clue.
Step 2: Check acuity and danger - fever/neutropenia, bleeding, DIC, leukostasis, hyperviscosity, TLS, spinal cord compression, renal failure, hypercalcemia.
Step 3: Use the smear - blasts, Auer rods, smudge cells, rouleaux, teardrops, dysplasia, basophilia, nucleated RBCs.
Step 4: Confirm lineage - flow cytometry for blasts/lymphocytes; SPEP/UPEP/free light chains for paraprotein; BCR-ABL1/JAK2/CALR/MPL when MPN suspected.
Step 5: Stage/risk-stratify after diagnosis - imaging, marrow biopsy, cytogenetics/molecular testing, organ function, performance status.

Plasma cell disorders

Plasma cells are terminally differentiated B cells that produce immunoglobulin. A plasma cell clone produces a monoclonal protein, often called an M protein or paraprotein. The clinical meaning of the M protein depends on tumor burden and end-organ damage. MGUS is common with aging and is asymptomatic; smoldering myeloma is higher burden without myeloma-defining events; symptomatic multiple myeloma causes organ injury. Waldenstrom macroglobulinemia is different: it is a lymphoplasmacytic lymphoma that secretes IgM and can cause hyperviscosity.

The classic myeloma organ-injury mnemonic is CRAB: hyperCalcemia, Renal injury, Anemia, and Bone lesions. Modern criteria also recognize myeloma-defining biomarkers, such as a very high marrow plasma cell burden, involved/uninvolved free light-chain ratio, or multiple focal lesions on MRI, because these predict near-term progression. For internal medicine pattern recognition, the key is to connect monoclonal protein to bone destruction, kidney injury, immune dysfunction, and hyperviscosity or amyloid phenomena.

Disorder	Core definition	Typical clues	Management concept
MGUS	M protein with low clonal plasma cell burden and no end-organ injury	Incidental protein gap, older age, no CRAB	Observation/risk-based follow-up
Smoldering myeloma	Higher tumor burden than MGUS without myeloma-defining organ damage	M protein and marrow plasma cells but no CRAB	Close monitoring or clinical-trial/risk-adapted therapy in selected patients

Disorder	Core definition	Typical clues	Management concept
Multiple myeloma	Clonal plasma cell disease with myeloma-defining event	CRAB, recurrent infections, lytic lesions, rouleaux, Bence Jones protein	Systemic therapy, bone protection, renal protection, transplant eligibility assessment
Solitary plasmacytoma	Single plasma-cell mass in bone or soft tissue	Localized pain or mass; limited marrow involvement	Local radiation, surveillance for progression
Waldenstrom macroglobulinemia	Lymphoplasmacytic lymphoma with IgM paraprotein	Hyperviscosity, neuropathy, bleeding, lymphadenopathy/splenomegaly	Treat symptomatic disease; plasmapheresis for hyperviscosity emergency
AL amyloidosis	Misfolded light chains deposit in organs	Nephrotic proteinuria, restrictive cardiomyopathy, hepatomegaly, macroglossia, neuropathy	Plasma-cell directed therapy; organ-specific supportive care

Key diagnostic tests in plasma cell disease

SPEP detects and quantifies a serum M spike but may miss light-chain-only disease. Serum immunofixation identifies the immunoglobulin type. UPEP/urine immunofixation detects urinary light chains, historically called Bence Jones proteins. Serum free light-chain testing is sensitive for light-chain disorders and helps risk stratification. Bone marrow biopsy quantifies clonal plasma cells and provides cytogenetics. Imaging has moved beyond plain skeletal survey in many settings; low-dose whole-body CT, PET/CT, or MRI can detect lytic lesions or focal marrow disease.

Complication map

Complication	Mechanism	Bedside association
Renal failure	Light-chain cast nephropathy, hypercalcemia, dehydration, infection, NSAIDs/contrast	High creatinine plus protein gap; urine dip may underestimate light-chain protein
Hypercalcemia	Osteoclast activation and bone resorption	Constipation, confusion, polyuria, dehydration, shortened QT
Bone pain/fracture	Lytic lesions and cortical destruction	Back/rib pain; vertebral compression; avoid assuming simple degenerative pain
Anemia	Marrow replacement and inflammatory suppression	Fatigue, dyspnea, normocytic anemia
Infection	Impaired normal immunoglobulin production	Recurrent sinopulmonary infections
Hyperviscosity	High circulating paraprotein, especially IgM	Headache, blurred vision, mucosal bleeding, neurologic symptoms

Clinical pearl: a protein gap is not a diagnosis. Total protein minus albumin suggests excess globulin, but monoclonality requires electrophoresis/immunofixation or related testing. Polyclonal hypergammaglobulinemia from inflammation, infection, liver disease, or autoimmune disease can also widen the gap.

Lymphadenopathy and lymphoma framework

Lymphomas are malignancies of lymphocytes or their precursors. The first clinical clue is often painless lymphadenopathy, but extranodal disease can occur in the GI tract, skin, CNS, testis, bone marrow, or other organs. A useful first split is Hodgkin lymphoma versus non-Hodgkin lymphoma. Hodgkin lymphoma is defined by Reed-Sternberg cells in the right inflammatory background and often spreads contiguously from one nodal region to the next. Non-Hodgkin lymphomas are a heterogeneous group of B-cell, T-cell, and NK-cell neoplasms whose behavior ranges from indolent to rapidly fatal without therapy.

Excisional lymph node biopsy is preferred when lymphoma is suspected because architecture matters. Fine-needle aspiration may identify metastatic carcinoma or infection, but it can miss lymphoma subclassification. After diagnosis, PET/CT and bone marrow assessment may help stage disease, but the diagnostic priority is tissue.

Feature	Hodgkin lymphoma	Non-Hodgkin lymphoma
Typical spread	Often contiguous nodal spread	Often noncontiguous; extranodal disease more common
Diagnostic cell	Reed-Sternberg cell in inflammatory background	Depends on subtype; flow/IHC define lineage
Age pattern	Bimodal: young adults and older adults	More common with increasing age; subtype-dependent
Common presentation	Painless cervical/supraclavicular nodes, mediastinal mass, B symptoms	Painless nodes, extranodal mass, marrow involvement, B symptoms
Treatment principle	Stage- and risk-adapted chemotherapy +/- radiation; high cure rates in many cases	Subtype-driven: watchful waiting to urgent immunochemotherapy/targeted therapy

Ann Arbor-style lymphoma staging

Stage	Meaning
I	One lymph node region or one extralymphatic site
II	Two or more nodal regions on the same side of the diaphragm, or localized extranodal extension with same-side nodes
III	Nodal regions on both sides of the diaphragm
IV	Diffuse/disseminated involvement of one or more extranodal organs such as liver, marrow, lung, or CNS
A/B	A means no systemic B symptoms; B means fever, drenching night sweats, or unexplained weight loss

Hodgkin lymphoma illness script

Hodgkin lymphoma often presents with painless lymphadenopathy, commonly cervical or supraclavicular, and may cause mediastinal widening. B symptoms include fever, drenching night sweats, and weight loss. Pruritus and alcohol-induced lymph node pain are classic but not necessary. Nodular sclerosis is common in young women and often involves the mediastinum. Mixed cellularity is more associated with older age, HIV, and EBV. Lymphocyte-depleted disease is less common and generally has worse prognosis.

Non-Hodgkin lymphoma high-yield subtypes

Subtype	Behavior	Key associations/clues
Follicular lymphoma	Indolent B-cell lymphoma	Painless waxing/waning nodes; t(14;18)/BCL2; may transform to DLBCL
Diffuse large B-cell lymphoma	Aggressive but potentially curable	Rapidly enlarging mass; extranodal disease; common adult NHL
Burkitt lymphoma	Very aggressive	MYC translocation; EBV association; jaw/facial mass in endemic form; abdominal mass in sporadic form; high TLS risk
Mantle cell lymphoma	Often aggressive/relapsing	Cyclin D1 overexpression; t(11;14); older men; GI polyposis can occur
Marginal zone/MALT lymphoma	Indolent	Chronic antigen stimulation; H. pylori gastric MALT, Sjogren, Hashimoto
Adult T-cell leukemia/lymphoma	Aggressive T-cell neoplasm	HTLV-1; Japan/Caribbean associations; skin lesions, hypercalcemia
Primary CNS lymphoma	Extranodal aggressive lymphoma	Immunosuppression/HIV; brain lesions; avoid steroids before biopsy if safe

Clinical trap: do not drain or empirically excise an unexplained persistent lymph node as if it were a simple abscess unless infection is clear. Lymphoma diagnosis depends on tissue architecture, immunophenotype, and sometimes cytogenetics. A poorly sampled node can delay classification and treatment.

Leukemia framework

Leukemias are malignant disorders of blood-forming cells. Acute leukemias are dominated by blasts and progress rapidly; chronic leukemias are more mature-cell disorders and often present more indolently. A CBC may show leukocytosis, normal WBC count, or leukopenia. The most dangerous presentations are febrile neutropenia, severe anemia, thrombocytopenic bleeding, leukostasis, DIC, and tumor lysis syndrome.

The source section begins acute leukemia by emphasizing that acute leukemias commonly present with anemia, infection, bleeding, splenomegaly/hepatomegaly/lymphadenopathy, bone pain, and sometimes CNS/testicular involvement. That symptom cluster follows directly from marrow replacement: normal erythropoiesis falls, granulocyte function falls, and platelet production falls, while malignant cells infiltrate organs.

Question	AML	ALL	CML	CLL
Cell maturity	Myeloid blasts	Lymphoblasts	Mature granulocytic lineage overproduction	Mature clonal B lymphocytes
Typical age	Adults, older adults	Children most common, also adults	Adults	Older adults
Classic clue	Auer rods; cytopenias; DIC in APL	Bone pain; lymphadenopathy; mediastinal mass in T-ALL	Very high WBC, basophilia, splenomegaly, Philadelphia chromosome	Asymptomatic lymphocytosis, smudge cells, hypogammaglobulinemia
Key test	Flow cytometry + marrow + cytogenetics/molecular	Flow cytometry + marrow +/- CNS evaluation	BCR-ABL1	Flow cytometry showing clonal B-cell population

Question	AML	ALL	CML	CLL
Treatment concept	Risk-adapted induction/consolidation; targeted agents depending mutations	Multi-agent therapy; CNS prophylaxis	Tyrosine kinase inhibitor	Observe if early/asymptomatic; targeted therapy if symptomatic/progressive

AML and APL

AML is a clonal myeloid precursor malignancy. Symptoms usually reflect cytopenias and tissue infiltration. Auer rods are needle-like azurophilic granules that support myeloid lineage. Diagnosis requires marrow/blood evaluation with flow cytometry, cytogenetics, and molecular testing because prognosis and therapy depend heavily on genetics. Acute promyelocytic leukemia is a special emergency subtype associated with PML-RARA and high risk of DIC. If APL is suspected clinically or morphologically, all-trans retinoic acid is started urgently while confirmatory testing proceeds because early hemorrhagic death is preventable.

ALL

ALL is a lymphoblast malignancy, most common in children but also seen in adults. It may present with fever, bone pain, hepatosplenomegaly, lymphadenopathy, testicular involvement, or CNS symptoms. T-ALL can produce an anterior mediastinal mass that may cause airway symptoms or superior vena cava syndrome. Treatment uses multi-agent chemotherapy with CNS prophylaxis; adult therapy and risk stratification depend on age, cytogenetics, MRD response, and Philadelphia chromosome status.

CML

CML is driven by BCR-ABL1, the Philadelphia chromosome fusion tyrosine kinase. It often presents with fatigue, weight loss, splenomegaly, early satiety, leukocytosis across granulocytic maturation stages, and basophilia. Leukocyte alkaline phosphatase is historically low in CML and high in leukemoid reaction, but modern diagnosis relies on BCR-ABL1 testing. Tyrosine kinase inhibitors transformed CML into a chronic, monitorable disease for many patients. Blast crisis resembles acute leukemia and requires urgent specialist therapy.

CLL/SLL

CLL is a mature clonal B-cell leukemia; SLL is the tissue-predominant counterpart. Many patients are discovered incidentally with persistent lymphocytosis. Smudge cells are fragile lymphocytes disrupted on smear. Clinical problems include lymphadenopathy, splenomegaly, autoimmune hemolytic anemia, immune thrombocytopenia, hypogammaglobulinemia, and recurrent infections. Early asymptomatic disease is commonly observed rather than immediately treated; treatment is triggered by symptoms, progressive cytopenias, bulky disease, or rapid progression.

Leukemoid reaction vs leukemia

Feature	Leukemoid reaction	CML/leukemia concern
Cause	Reactive infection/inflammation, stress, steroids, severe illness	Clonal hematologic malignancy
Smear	Mature neutrophilia with left shift; toxic granulation/Dohle bodies may appear	Granulocytes at multiple stages, basophilia/eosinophilia, blasts depending disease
LAP score	Historically high	Historically low in CML
Spleen	May be normal or mildly enlarged	Splenomegaly common in CML/MPNs
Confirmatory test	Treat underlying cause and trend counts	Molecular/flow/marrow testing such as BCR-ABL1

Myeloproliferative neoplasms and myelodysplastic syndromes

Myeloproliferative neoplasms are clonal marrow disorders that overproduce relatively mature myeloid cells. The classic BCR-ABL1-negative MPNs are polycythemia vera, essential thrombocythemia, and primary myelofibrosis. They share risks of thrombosis, bleeding, splenomegaly, constitutional symptoms, and transformation to acute leukemia. Myelodysplastic syndromes are clonal disorders of ineffective hematopoiesis: the marrow may be cellular, but the peripheral blood shows cytopenias because cells mature abnormally and die early.

Disease	Dominant abnormality	Clinical clues	Key testing/management concept
Polycythemia vera	Increased RBC mass, often with leukocytosis/thrombocytosis	Headache, pruritus after hot shower, erythromelalgia, thrombosis, splenomegaly	JAK2 common; low EPO; phlebotomy/aspirin/cytoreduction by risk
Essential thrombocythemia	Sustained thrombocytosis	Thrombosis, bleeding, erythromelalgia, incidental high platelets	JAK2/CALR/MPL; rule out reactive thrombocytosis
Primary myelofibrosis	Marrow fibrosis and extramedullary hematopoiesis	Massive splenomegaly, constitutional symptoms, teardrop cells, dry tap	JAK2/CALR/MPL; supportive/cytoreductive/JAK inhibitor/transplant in selected
MDS	Cytopenias with dysplasia	Older adult, macrocytosis, neutropenia/infections, thrombocytopenia/bleeding	Marrow dysplasia, cytogenetics; supportive care, hypomethylating agents, transplant in selected

Oncologic and hematologic emergencies

A hematologic malignancy question is often really an emergency recognition question. Stabilization comes before perfect classification. The highest-yield emergencies are neutropenic fever, leukostasis, tumor lysis syndrome, DIC in APL or sepsis-associated malignancy, hypercalcemia of malignancy, hyperviscosity, spinal cord compression, and superior vena cava syndrome.

Emergency	Mechanism	Clues	Immediate management concept
Febrile neutropenia	Low neutrophils impair bacterial/fungal containment	Fever with ANC <500 or expected decline	Cultures and prompt antipseudomonal antibiotics; do not wait for source
Tumor lysis syndrome	Rapid cell lysis releases K, phosphate, nucleic acids; calcium falls	Hyperkalemia, hyperphosphatemia, hypocalcemia, hyperuricemia, AKI	IV fluids, urate-lowering therapy, electrolyte control, dialysis if severe
Leukostasis	Very high blasts/WBC impair microcirculation	Dyspnea, hypoxemia, headache, confusion, visual symptoms	Urgent cytoreduction with specialist input; avoid unnecessary RBC transfusion if hyperviscous
Hyperviscosity	Paraprotein thickens blood	Headache, visual changes, mucosal bleeding, neurologic symptoms	Plasmapheresis for severe symptoms; then disease-directed therapy
Spinal cord compression	Epidural tumor or vertebral collapse	Back pain, weakness, sensory level, bowel/bladder dysfunction	Urgent MRI, glucocorticoids when appropriate, radiation/surgery consult
DIC	Systemic coagulation activation consumes platelets/fibrinogen	Bleeding + thrombosis, prolonged PT/PTT, low fibrinogen, high D-dimer	Treat trigger; support with platelets/cryo/FFP if bleeding/procedure

Tumor lysis syndrome numeric anchors

Classic laboratory tumor lysis includes rising potassium, phosphate, uric acid, and falling calcium. High-risk diseases include Burkitt lymphoma, lymphoblastic leukemia/lymphoma, AML with high tumor burden, and bulky chemosensitive tumors. Prevention matters: aggressive hydration and urate-lowering therapy are easier than rescuing severe hyperkalemia and acute kidney injury after lysis begins.

Breast cancer and solid oncology bridge

Although this chapter focuses on hematologic malignancy, the uploaded source briefly reviews breast cancer. The reason to include a short bridge is that general oncology principles are often tested across cancers: risk factors, screening versus diagnosis, biopsy, receptor testing, staging, and treatment sequencing. Breast cancer is not diagnosed by mammography alone when a palpable mass is present; diagnostic imaging and tissue sampling are needed. Receptor status - estrogen receptor, progesterone receptor, and HER2 - guides systemic therapy. Lymph node status and tumor size influence staging and adjuvant treatment. In the same way, hematologic malignancies require immunophenotype, cytogenetics, and molecular markers before therapy is finalized.

General oncology step	Breast cancer example	Heme-onc parallel
Find the lesion	Diagnostic mammogram/ultrasound for mass	CBC/smear/imaging identifies abnormal compartment
Get tissue	Core needle biopsy	Excisional node biopsy or marrow biopsy/flow
Classify biology	ER/PR/HER2, grade	Flow markers, cytogenetics, mutations, clonality
Stage extent	Tumor/nodes/metastasis	Ann Arbor, marrow involvement, PET/CT, organ injury
Treat safely	Surgery/systemic/radiation sequence	Stabilize emergencies, then subtype-specific therapy

Diagnostic algorithms

Algorithm 1: Adult with painless lymphadenopathy

1. Screen for red flags: node >2 cm, supraclavicular location, firmness/fixation, generalized nodes, B symptoms, hepatosplenomegaly, abnormal CBC, immunosuppression.
2. If infection is likely and short-lived, treat or observe briefly with reassessment; do not repeatedly give antibiotics for persistent unexplained nodes.
3. Obtain CBC with differential, smear, HIV testing when appropriate, and targeted infectious/autoimmune tests based on exposure history.
4. Persistent high-risk node: excisional biopsy preferred. Preserve tissue for flow cytometry, immunohistochemistry, cytogenetics, and microbiology if infection remains possible.
5. After lymphoma diagnosis: PET/CT or CT staging, marrow evaluation when indicated, LDH, hepatitis/HIV testing before immunochemotherapy, fertility and vaccination planning when appropriate.

Algorithm 2: Monoclonal protein or protein gap

1. Confirm monoclonality: SPEP with immunofixation, serum free light chains, and urine studies if suspicion persists.
2. Assess end-organ injury: calcium, creatinine/eGFR, CBC, quantitative immunoglobulins, urine protein, bone pain/fracture history.
3. Image the skeleton if myeloma suspected; use modern cross-sectional imaging when available rather than relying only on plain films.
4. Bone marrow biopsy when myeloma, amyloid, or another clonal plasma-cell disorder is suspected.
5. Separate MGUS/smoldering from symptomatic disease. Treat end-organ injury and emergencies: hypercalcemia, renal failure, spinal cord compression, infection, hyperviscosity.

Algorithm 3: Leukocytosis with possible leukemia

1. Check symptoms: fever, infection, bleeding, dyspnea, neurologic symptoms, visual changes, splenomegaly, bone pain.
2. Review CBC differential and smear: blasts, basophilia, left shift, lymphocytosis, smudge cells, dysplasia.
3. If blasts or unstable patient: urgent hematology; evaluate DIC/TLS; avoid delays for outpatient workup.
4. Use flow cytometry to define lineage. Use molecular/cytogenetic testing for BCR-ABL1, PML-RARA, FLT3/NPM1/IDH and other disease-specific markers as directed.
5. Treat emergency physiology first: broad-spectrum antibiotics for neutropenic fever, ATRA for suspected APL, hydration/TLS prevention, cytorreduction for leukostasis.

High-yield comparison tables

Finding	Most likely disease pattern	Why it matters
Rouleaux on smear	Multiple myeloma or other paraproteinemia	Excess protein reduces RBC repulsion, stacking cells
Reed-Sternberg cells	Hodgkin lymphoma	Diagnostic in appropriate tissue context
Auer rods	AML	Myeloid blasts; APL concern if promyelocytes/DIC
Smudge cells	CLL	Fragile mature lymphocytes disrupted on smear
Basophilia plus very high WBC	CML	BCR-ABL1-driven granulocytic proliferation
Teardrop cells	Myelofibrosis or marrow infiltration	Distorted marrow architecture/extramedullary hematopoiesis
Lytic punched-out bone lesions	Multiple myeloma	Osteoclast activation and plasma-cell bone disease
IgM hyperviscosity	Waldenstrom macroglobulinemia	Large pentameric IgM thickens blood

Disease	Key immune marker / genetic clue	Classic association
APL	PML-RARA	DIC; treat urgently with ATRA
CML	BCR-ABL1	Splenomegaly, basophilia, TKI response
Burkitt lymphoma	MYC translocation	Starry-sky histology, EBV association, jaw/abdominal mass, TLS

Disease	Key immune marker / genetic clue	Classic association
Follicular lymphoma	BCL2 overexpression from t(14;18)	Indolent waxing/waning nodes; transformation risk
Mantle cell lymphoma	Cyclin D1/t(11;14)	Older men, GI involvement, aggressive course
CLL	Clonal mature B cells, often CD5-positive	Smudge cells, autoimmune cytopenias, hypogammaglobulinemia
PV/ET/MF	JAK2/CALR/MPL patterns	Thrombosis, splenomegaly, variable cytos

Practice questions

Question 1. An older adult has back pain, creatinine elevation, calcium 12.4 mg/dL, normocytic anemia, and total protein much higher than albumin. What disease pattern links these findings?

Answer: Multiple myeloma. The unifying mechanism is a clonal plasma-cell disorder producing monoclonal protein with CRAB end-organ injury: hypercalcemia from bone resorption, renal injury from light chains/hypercalcemia/dehydration, anemia from marrow infiltration/inflammation, and bone pain from lytic lesions.

Question 2. A young adult has painless cervical lymphadenopathy, night sweats, and an excisional node biopsy showing Reed-Sternberg cells. What staging symptom suffix applies, and why does excisional biopsy matter?

Answer: The B suffix applies because night sweats are a B symptom. Excisional biopsy preserves lymph node architecture and allows immunophenotyping, which is important for lymphoma classification.

Question 3. A patient with suspected acute leukemia has bleeding gums, prolonged PT/PTT, low fibrinogen, and abnormal promyelocytes. What is the emergency diagnosis and immediate disease-directed medication?

Answer: Acute promyelocytic leukemia with DIC. Start all-trans retinoic acid immediately while confirmatory PML-RARA testing proceeds.

Question 4. A patient with lymphocytosis has smudge cells but no anemia, thrombocytopenia, bulky nodes, or symptoms. What is the likely diagnosis and broad management concept?

Answer: CLL. Many asymptomatic early-stage patients are observed rather than treated immediately; therapy is triggered by symptomatic or progressive disease.

Question 5. A patient has IgM monoclonal gammopathy, headaches, mucosal bleeding, blurred vision, lymphadenopathy, and splenomegaly. What emergency therapy may be needed before definitive disease-directed treatment?

Answer: Waldenström macroglobulinemia with hyperviscosity syndrome. Symptomatic hyperviscosity is treated with plasmapheresis, followed by disease-directed therapy.

Question 6. A patient with massive splenomegaly, fatigue, very high WBC, basophilia, and granulocytes at many maturation stages has which key molecular test?

Answer: BCR-ABL1 testing for CML.

Question 7. A patient with rapidly enlarging abdominal mass from Burkitt lymphoma starts chemotherapy and develops hyperkalemia, hyperphosphatemia, hypocalcemia, hyperuricemia, and AKI. Name the syndrome and prevention strategy.

Answer: Tumor lysis syndrome. Prevention includes risk stratification, aggressive hydration, monitoring, and urate-lowering therapy when indicated.

Question 8. A persistent supraclavicular lymph node is firm and painless in an adult with weight loss. What is the most appropriate diagnostic step?

Answer: Tissue diagnosis, preferably excisional lymph node biopsy when lymphoma is suspected, rather than repeated empiric antibiotics or reassurance.

Common pitfalls

- Assuming a normal or low WBC rules out acute leukemia. Acute leukemia can present with leukopenia, normal WBC, or leukocytosis; the smear and marrow are decisive.
- Treating persistent lymphadenopathy with repeated antibiotics without a tissue plan. Persistent, firm, supraclavicular, generalized, or B-symptom-associated nodes need malignancy evaluation.
- Calling every monoclonal protein multiple myeloma. MGUS and smoldering disease lack myeloma-defining organ injury, but require risk-based follow-up.
- Missing APL-associated DIC. Suspected APL is a hematologic emergency because early ATRA reduces fatal bleeding risk.
- Focusing on exact chemotherapy names before stabilizing oncologic emergencies. Hypercalcemia, TLS, leukostasis, hyperviscosity, neutropenic fever, and cord compression are immediate priorities.
- Using screening rules for symptomatic patients. Screening applies to asymptomatic risk groups; symptoms require diagnostic evaluation.

Graph-RAG extraction map

The following map is designed for concept-node extraction. Each node can hold disease definitions, numeric anchors, diagnostic tests, complications, and source-context links. Edges should be explicit because many exam questions depend on relationship activation rather than isolated recall.

Node	Class/subclass	Core properties	High-yield edges
Multiple myeloma	Heme malignancy -> plasma cell neoplasm	Clonal plasma cells; M protein; CRAB; lytic lesions; renal injury; rouleaux	causes -> hypercalcemia, renal failure, anemia, infections; diagnosed_by -> SPEP/UPEP/free light chains/marrow; differentiated_from -> MGUS
MGUS	Plasma cell premalignant disorder	M protein, low plasma cell burden, no CRAB	progresses_to -> myeloma; managed_by -> observation
Waldenstrom macroglobulinemia	Lymphoplasmacytic lymphoma / IgM paraprotein disorder	IgM monoclonal protein; hyperviscosity; neuropathy; bleeding	causes -> hyperviscosity; treated_emergently_by -> plasmapheresis
Hodgkin lymphoma	Lymphoid malignancy	Reed-Sternberg cells; contiguous spread; B symptoms; mediastinal disease	diagnosed_by -> excisional node biopsy; staged_by -> Ann Arbor/PET
Diffuse large B-cell lymphoma	NHL -> aggressive B-cell lymphoma	Rapid mass; extranodal disease; potentially curable	treated_by -> immunochemotherapy; complication -> TLS
Burkitt lymphoma	NHL -> high-grade B-cell lymphoma	MYC; starry-sky; EBV; jaw/abdominal mass; high proliferation	high_risk_for -> TLS
AML	Leukemia -> acute myeloid	Blasts; Auer rods; cytopenias; adult predominance	emergency_variant -> APL; diagnosed_by -> flow/marrow/cytogenetics
APL	AML subtype	PML-RARA; promyelocytes; DIC	treated_immediately_by -> ATRA; causes -> hemorrhage/DIC
ALL	Leukemia -> acute lymphoblastic	Lymphoblasts; children; bone pain; CNS/testicular disease	requires -> CNS prophylaxis; subtype -> T-ALL mediastinal mass
CML	MPN / chronic leukemia	BCR-ABL1; basophilia; splenomegaly	treated_by -> tyrosine kinase inhibitor; transforms_to -> blast crisis
CLL	Chronic lymphoid leukemia	Older adult lymphocytosis; smudge cells; hypogammaglobulinemia	complicates -> autoimmune hemolysis, infections; managed_by -> observation if asymptomatic
Tumor lysis syndrome	Oncologic emergency	K high, phosphate high, uric acid high, calcium low, AKI	triggered_by -> chemo/high tumor burden; prevented_by -> hydration/urate lowering

Source-guided coverage notes

This chapter uses the uploaded hematology/oncology segment as a coverage guide. The source segment includes general oncology epidemiology, prevention/screening, staging and therapy principles, a breast cancer bridge, MGUS and multiple myeloma, Waldenstrom macroglobulinemia, Hodgkin and non-Hodgkin lymphoma, and the beginning of acute leukemia. The Medvale version expands those signals into an original integrated framework for hematologic malignancies and graph-RAG extraction.

Selected references for clinical cross-checking

- National Cancer Institute. Adult Acute Myeloid Leukemia Treatment (PDQ) - Health Professional Version. Updated 2025.
- National Cancer Institute. Plasma Cell Neoplasms (Including Multiple Myeloma) Treatment (PDQ) - Health Professional Version. Updated 2025.
- National Cancer Institute. Adult Hodgkin Lymphoma Treatment (PDQ) - Health Professional Version. Updated 2025.
- National Comprehensive Cancer Network. NCCN Guidelines resources for AML and Multiple Myeloma. Accessed for current guideline context.
- Uploaded source segment: heme3.pdf, 20-page hematology/oncology extract supplied by the user, used as a topic coverage signal only.